

# Analysis of Contemporary Economic Issues

## Guidance 7: Working from home and COVID-19: labor-market and housing market impact

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2026



**SOLVAY BRUSSELS SCHOOL  
ECONOMICS & MANAGEMENT**

# Outline

1 Working from Home: Labor & Housing Market Impact

2 Questions

# Working From Home (WFH): Definition & Context

## Definition

**Telework:** Work from home (WFH) using digital tools.

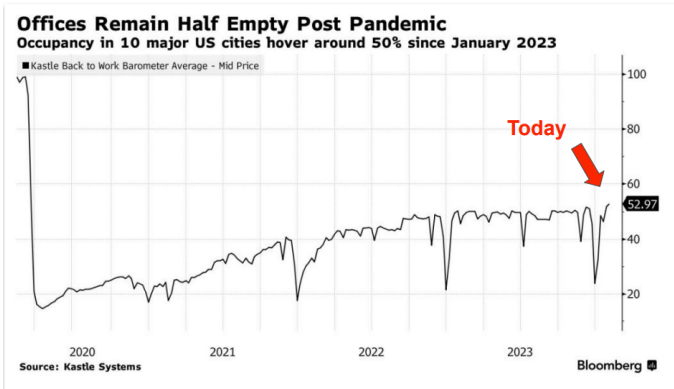
- Can be Full time or "Hybrid" → post-Covid reality.

## The COVID-19 Shock:

- Started as a temporary exogenous shock (Lockdown 2020-2023).
- **Current Status:** Telework persists, and may still be the case in the future.
- **Consequence:** Permanent long-term structural changes → hysteresis.
- **Why ?:** Workers value flexibility and saved commuting time, and firms realized they could save on office space costs.

# The Persistence of WFH: Office Occupancy

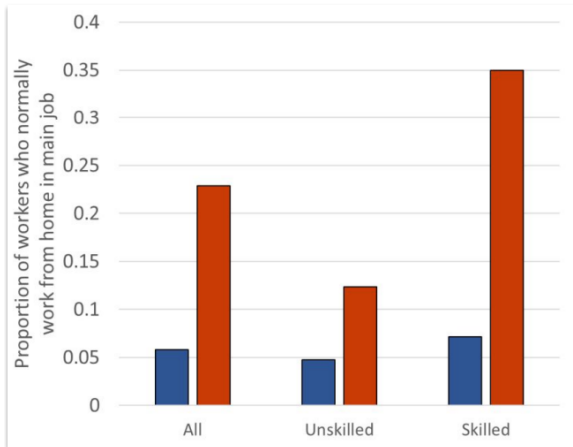
**Observation:** Offices remain half empty post-pandemic.



## Timeline:

- **Pre-Pandemic:**  $\approx$  90-100% occupancy.
- **First Lockdown (2020):** Sharp drop to  $< 20\%$ .
- **Today (2023+):** Hovering around **50%** in major US cities. But it's the new "normal", not a recovery.

# Who Works from Home? (Skills Gap)



## Inequality in WFH

**Telework is easier for skilled workers than unskilled workers.**

- **Skilled workers:** Massive increase in WFH proportion (2019 vs 2023).
- **Unskilled workers:** Moderate increase.

**Why ?:**

- **Skilled workers** produce outputs that are digital and transmittable.
- **Unskilled workers** produce outputs that require physical presence.

# Impact on Commuting Distance

- **Before 2020:** Only 5% of EU workers worked from home systematically.
- **Post-Pandemic:** Workers live further from their employers. When you don't need to commute daily, your residential location becomes decoupled from your job location.
- **Most affected sectors:** IT, Finance & Insurance and workers hired after March 2020 (because recruited under remote policies). Where work is most digitizable.

**Example:** Someone working for a Brussels-based consultancy could now plausibly live in Namur, Liège, or even across the border , commuting in only once or twice a week.

| Metric                                      | 2019             | 2023                    |
|---------------------------------------------|------------------|-------------------------|
| Mean distance (workplace ↔ residence)       | 10 miles (16 km) | <b>27 miles (43 km)</b> |
| % of Workers living > 50 miles (80 km) away | 0.8%             | <b>5.5%</b>             |

# How Effective is WFH?

**Seminal Paper:** Bloom et al. (2015) - *"Does working from home work? Evidence from a Chinese experiment"*

- **Context:** Study on Call Center employees (9 months).
- **Research Question:** Does WFH = "Shirking From Home"?
- (*Shirking* = avoiding/neglecting duty).

## Main Results (Field Experiment)

- **Output:** +13% performance increase.
- **Working Time:** +9% (fewer breaks, sick days).
- **Productivity:** +4% (quieter/convenient environment).

**VERDICT: EFFECTIVE**

# Field Experiment and Validity

## 2 types of experiments:

- **Lab experiment:** Artificial environment, hiring a group of people. The goal is to reproduce some features from real life.
- **Field experiment:** Interfering into people's life (Bloom et al.). Paying volunteers to participate and ask them to act as usual.

## Validity:

- **Internal Validity:** Is the causal relationship trustworthy (i.e not influenced by other factors or variables) ? Yes, because randomization ensures that any difference in outcomes between the WFH and office groups is caused by the treatment , not by pre-existing differences between the workers. No randomization → selection problem.
- **External Validity:** Can results be generalised to a larger context ? Complicated.
  - What about highly skilled workers (vs call center)?
  - Geographic extension?
  - Heterogeneity of workers' preferences?



# Impact of WFH on Cities

## Chain of Events:

- ① WFH → Less need to commute daily.
- ② Change in housing preferences:
  - Closer to nature.
  - Need for Home Office space → Bigger houses needed.
  - Typically found in suburbs or smaller cities, not in dense urban centers.
- ③ **Result:** Move away from city centers.

## Local Consumption Services (LCS):

- Services available in city centers (produced by low-pay workers). They cluster around office areas because their business model depends on foot traffic from office workers.
- WFH reduces spending on LCS near workplaces (Unemployment risk for service workers).
- When skilled knowledge workers shift to WFH (benefiting from flexibility and cost savings), the economic consequences cascade down to low-wage service workers who lose income or jobs.

# The Monocentric City Model

Providence, Rhode Island, in 1896:



**The "mill town" pattern:** everyone commutes to the center!  
Even today: **remarkably robust!**

## Concept:

- Most jobs located at a **CBD** (Central Business District), and workers live at various distances from it and commute in.
- "Mill town pattern": Everyone commutes to the center.
- **Examples:** US Cities, Paris, Brussels (Ring/La Cambre).

## LCS Spending near work:

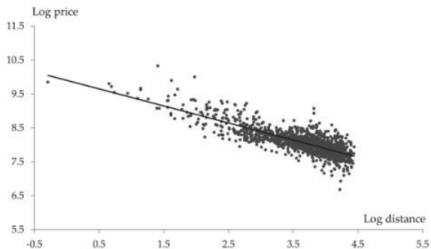
- UK: 4.9% of income.
- US: 12.5% ( $\approx \$7,904$ ).

# Effect on Housing Market

**Theory:** Land price usually decreases with distance from CBD.

## Paris (2012)

Land prices in Paris in 2012 (Combes et al., 2019):



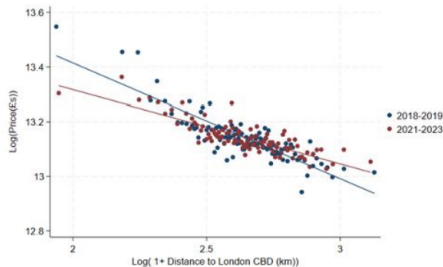
**The most striking feature:** land price ↓ with distance from the CBD

Living far from the CBD means higher commuting costs (time and money), so people are only willing to live far away if land is cheaper to compensate.

## London (Post-Pandemic Shift)

**Demand for proximity to the CBD:**

703,247 house sales 2018-2019 versus 2021-2023:



When workers telework, proximity to the CBD becomes less valuable, so prices near the center face downward pressure while prices in the periphery face upward pressure

# Effect on Housing Market

**Brueckner et al. (2023) Findings:** Study on the impacts of WFH on the housing market in the US both from intercity and intracity perspectives.

- **Intercity perspective:** WFH enables workers to keep high-paying jobs in productive cities while relocating to cheaper areas. This creates new housing demand in those cheaper areas, pushing up their prices and rents. So WFH makes previously affordable areas less affordable, as wealthier remote workers move in.
- **Intracity perspective:** Within cities, WFH flattens the house-price gradient. The price premium for living close to jobs weakens because job access matters less when you work remotely.
- **New spatial hedonic equilibrium:** Housing prices now reflect a different ranking of what matters, workers became less sensitive to distance between their home and CBD.
- Hedonic pricing theory says that the price of a good reflects its bundle of characteristics: size, number of rooms, neighborhood quality, school quality, and crucially, access to jobs.

# Any Questions?

## Clicker Questions

**1. (UA) When comparing global patterns of Foreign Direct Investment (FDI) and migration, which of the following statements are accurate? (Select all that apply)**

- A. Horizontal FDI mainly occurs between industrialized countries and represents the dominant form of direct investment.
- B. Vertical FDI is primarily driven by the search for lower production costs in developing countries.
- C. Just like international trade in goods, global migration flows are mostly bilateral (balanced back-and-forth movements between countries).
- D. Economic theory suggests that international trade can act as a substitute for labor migration by raising living standards in the origin countries.

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**2. (MA) In the context of the European Green Deal and the transition to climate neutrality, which of the following statements regarding policy instruments and economic goals are accurate? (Select all that apply)**

- A. Reaching the updated 2030 target of a 55% reduction in GHG emissions requires an annual emission reduction of 2.7%, which more than doubles the historical rate.
- B. A Carbon Tax sets a strict maximum volume of emissions allowed, whereas an Emission Trading Scheme (ETS) guarantees a fixed and predictable price for investors.
- C. Decoupling economic growth from resource use is one of the three key pillars of the European Green Deal.
- D. To manage carbon leakage risks—where companies might relocate to countries with lower carbon prices—the EU can use mechanisms like free ETS allowances or a Carbon Border Tax.



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**3. (UA) Belgium is ranked as a top 5 innovator in the EU with high R&D spending (3.4% of GDP). However, this does not translate into significant labor productivity gains. According to the presentation, what is the primary reason for this paradox?**

- A. R&D investments are almost exclusively directed towards the agricultural sector, which has low productivity growth.
- B. There is a severe "diffusion gap" where innovations, such as AI, are heavily concentrated in large enterprises but fail to penetrate SMEs.
- C. The National Bank of Belgium's macro-prudential policies strictly limit the creation of new startups.
- D. Belgium suffers from an extremely low rate of tertiary education, preventing the workforce from understanding new technologies.

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**4. (UA) The presentation highlights that Belgium suffers from low incentives for job mobility. Which structural factor is identified as a primary driver of this specific issue?**

- A. A massive surplus of STEM graduates that saturates the high-wage job market.
- B. Rigid wage-setting mechanisms (such as the Law of 1996) that result in a weak gap between median and high wages (a low D9/D5 inter-decile ratio).
- C. An excessively high job vacancy rate in Wallonia that disincentivizes workers from moving to Flanders.
- D. A strict limitation on unemployment benefits to a maximum of 6 months.

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**5. (Open-ended) The European Green Deal relies heavily on both market mechanisms and rigorous policy evaluation. Briefly explain the core mechanism of the EU Emissions Trading System (ETS). Name two new sectors that the European Commission proposes to include either in the existing ETS or in a separate, newly created ETS.**

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**5. (Open-ended) The European Green Deal relies heavily on both market mechanisms and rigorous policy evaluation. Briefly explain the core mechanism of the EU Emissions Trading System (ETS). Name two new sectors that the European Commission proposes to include either in the existing ETS or in a separate, newly created ETS.**

### **Expected Answer:**

**ETS Mechanism:** It puts a price on carbon and lowers the overall cap on allowable emissions every year.

**New sectors:** The Commission proposes including shipping emissions in the main ETS, and creating a separate new ETS for fuel distribution, road transport, and buildings. (Students must name at least two).

### 6. (UA) In the context of the course's conceptual framework, how is the "greenium" specifically defined?

- A. The initial wealth gap between "patient" and "impatient" investors.
- B. The phenomenon where the expected yield of a green asset is lower than the expected yield of a brown asset.
- C. A government subsidy given to companies to offset the financial costs of the ecological transition.
- D. The extra yield demanded by investors to compensate for the technological risks associated with sustainable projects.



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**7. (UA) According to the experiment conducted in a Chinese call center by Bloom et al. (2015), what was the main impact of working from home (WFH) on employee productivity and career paths?**

- A. WFH led to a 13% performance decrease due to numerous home distractions.
- B. Performance increased by 13%, primarily due to more minutes worked per shift (fewer breaks and sick days) and a quieter work environment.
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## Clicker Questions

**8. (MA) Based on the presentation regarding the impact of WFH on the labor market, which statements correctly describe the evolution of worker preferences and the post-COVID reality? (Select all that apply)**

- A. The persistence of telework after the pandemic is an example of "hysteresis," where an exogenous shock initially perceived as temporary leads to permanent, long-term structural changes.
- B. On average, employees attach a high monetary value to the option of teleworking, equivalent to a salary increase of about 8%.
- C. The hybrid work model (e.g., targeting 2 days a week at home) has established itself as a persistent new norm.
- D. The transition to telework benefited all types of professions perfectly equally, erasing spatial disparities between skilled and unskilled labor.

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### Expected Answer:

**Main components:** GDP (economic size) acts as an attractive force that increases trade flows, while distance acts as a friction that decreases them.

**Non-geographical distance:** Differences in language, lack of shared colonial history, different legal systems, or different currencies (Students must name at least two).

### 10. (UA) What is identified as a major risk or disadvantage associated with the implementation of the Carbon Border Adjustment Mechanism (CBAM)?

- A. It will force importers to pay a much higher carbon price than domestic producers pay under the ETS.
- B. It could lead to "Resource Shuffling," where exporting countries send their "cleanest" production to the EU while redirecting their "dirty" production to other markets.
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